

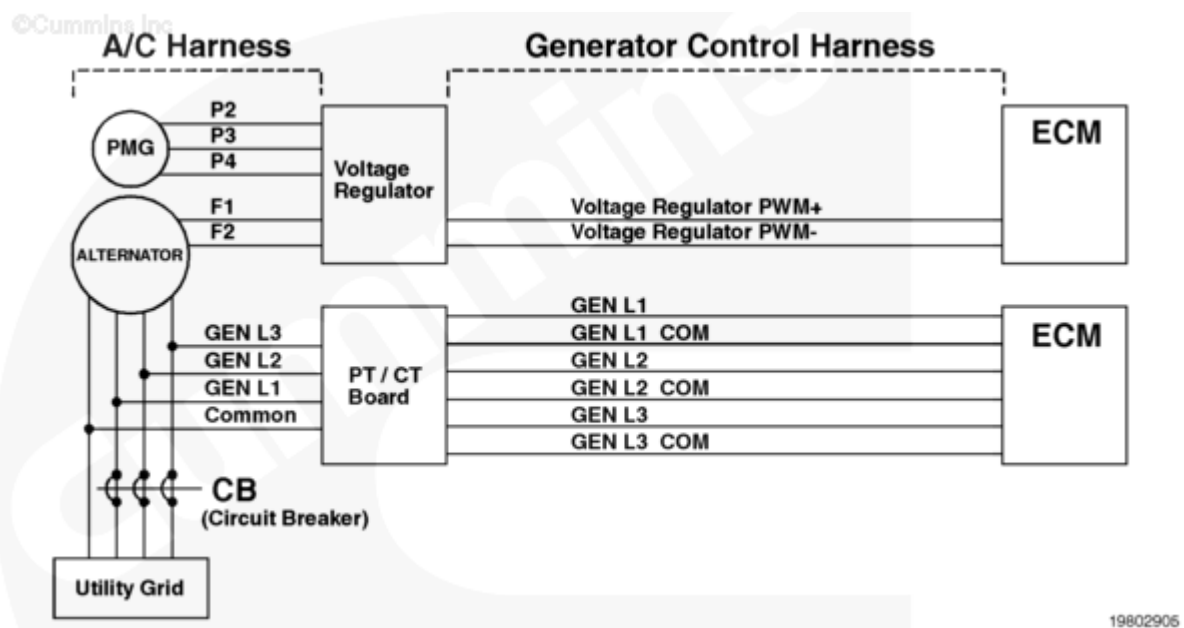
Fault Code: 1461

Loss of Electric Field Due to Reverse KVAR - Condition Exists

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Overview

Codes	Reason	Effect
Fault Code: 1461 PID(P): SPN: FMI: Lamp: Shutdown SRT:	Loss of field (electric) due to reverse kVAR.	Generator set will shut down.



Generator Circuit

Circuit Description

Under normal operating conditions, the generator set connects to the bus. The generator set is producing power, and when it connects to the bus, power is added to the main electric bus. When some events occur, a condition can occur when the generator set is no longer producing power for the electric bus, but rather is drawing power from the electric bus. This condition is called reverse kilovolt-ampere reduction (kVAR). Sometimes a reverse kVAR condition can cause the alternator to lose its electric field.

This fault code is used by the engine control module to tell the operator that the engine control module has detected a loss of electric field in the alternator due to a reverse kVAR condition.

Component Location

Reference Section E for location of the engine control module card cage.

Reference customer/facility/installation documentation for the location of the generator set circuit breaker and interface with the electric bus.

Shoptalk

The possible failure mode is perhaps due to power factor correction capacitors or other power sources feeding kVAR into the generator set.

Beyond a certain threshold, reverse kVAR can lead to voltage output instability and pole slipping due to the fact that the alternator becomes self-excited.

Check load-sharing lines for proper connection.

Refer to Troubleshooting Fault Code t05-1461. (</qs3/pubsys2/xml/en/procedures/01/01-t05-1461.html>)